

Rakey | Ruoqing Yang

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XR and HCI researcher designing and developing immersive systems,
conducting mixed-methods studies of perceptual and cognitive experience.

PUBLICATIONS

- **R. Yang** and C. Greenhalgh. “Comparing Experience Intensity of AR and VR for Contrasting Phobia Stimuli.” *Virtual Reality*, 2026.
- S. Jin, **R. Yang**, W. Tong, and L. H. Lee. “AIs or Humans in Luxury Shopping? A Design Study of VR Shopping Assistants.” *2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1155–1156, 2026.
- S. Jin, **R. Yang**, T. Bohn, W. Tong, and L. H. Lee. “Comparing Assistant Configurations in Simulated VR Luxury Shopping: A Mixed-Methods Study.” *Virtual Reality*, under review, 2026.

EDUCATION

- **University of Nottingham** 2023–2024
MSc, Human-Computer Interaction (Distinction) Nottingham, England
- **Ningbo University** 2018–2022
BSc, Logistics Management Ningbo, China

RESEARCH EXPERIENCE

- **Comparing Experience Intensity of AR and VR for Contrasting Phobia Stimuli** 2024–2025
MSc Dissertation · University of Nottingham Nottingham, England
 - Built matched VR and AR exposure environments in Unity to compare the experience intensity of spider and contamination stimuli across immersive environments.
 - Designed and conducted a headset-based AR/VR study (n=31), collecting self-report, behavioural, and interview data.
 - Found that VR elicited higher experience intensity overall, while AR performed comparably and offered greater ecological validity in the spider scenario, suggesting a staged AR-then-VR approach.
- **User Perception of Different Shopping Assistants in VR Luxury Store** 2025–2026
Research Collaboration · University of Cambridge Cambridge, England
 - Co-designed a multi-user PC-VR luxury store featuring four shopping-assistant modes: 3D avatar, webcam, voice, and AI.
 - Designed and conducted a real-time PC-VR study (n=20), examining workload, trust, preference, and social behaviour across four assistant modes.
 - Found that human assistants received better preference ratings and identified a realism-immersion trade-off; participant feedback also highlighted assistance timing and user agency.

PROJECTS

- **One’s Own Room – Adaptive WebXR Environment** [\[Github\]](#)
 - Designed a personalised WebXR space adapting to users’ affective state to support reflection and restoration.
 - Built an adaptive pipeline combining GenAI 3D world, affect-based atmosphere adjustment, and real-time voice interaction.
- **Virtual Exposure Therapy Application** [\[Github\]](#)
 - Extended the MSc research prototype into a VR/AR exposure-therapy application.
 - Enhanced the application with passthrough switching and hand-tracked touch-and-grab interactions for controller-free use.

PROFESSIONAL EXPERIENCE

- **AI Design Specialist** 01/2024–07/2024
O Shaped · Internship London, England
 - Built AI-avatar and generative-video workflows for educational content, reducing production time by 40%.
 - Iterated avatar appearance, behaviour, and speech based on user feedback, improving clarity and usability.

SKILLS

XR Development: Unity (C#) - VR/AR/MR (Passthrough, Hand/Eye Tracking), WebXR (Three.js, WebGL)

Research Methods: Mixed Methods (Quant/Qual), Research-through-Design, Co-design, Experimental Design, User Studies

Data Analysis: Quant - R, SPSS, Python; Qual - Qualitative Coding, Thematic Analysis

Design: Figma, Blender, Adobe Creative Suite **Languages:** English, Mandarin, Taiwanese Hokkien

AWARDS & ACTIVITIES

Presenter, GOSIM AI Spotlight, Paris (2026); Best AI Use Prize, RealityX AI × XR Hackathon (2026). First Prize, Cambridge EduX Hackathon (2025); Winter School - **Artificial Intelligence and Machine Learning**, University of Oxford (2022).